

CHEBOYGAN, MI, USA

WMO: 720321

Lat: 45.654N

Lon: 84.519W

Elev: 194

StdP: 99.02

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.0	-17.7	-24.2	0.4	-20.4	-21.2	0.6	-16.9	11.8	-5.9	10.9	-4.9	1.4	220	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.4	20.9	27.7	20.1	26.3	19.3	22.5	27.0	21.6	25.8	20.6	24.5	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.2	24.9	20.1	15.2	23.9	19.1	14.2	22.9	67.1	26.5	63.6	26.1	60.4	24.9	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.5	DB	-25.9	32.6	4.5	1.4	-29.2	33.6	-31.8	34.5	-34.3	35.3	-37.6	36.3	(4)
(5)			WB	-26.2	24.5	4.4	0.6	-29.4	24.9	-32.0	25.3	-34.4	25.6	-37.7	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-6.8	-6.9	-2.2	4.2	10.7	16.0	19.4	18.6	15.0	8.7	2.5	-3.0		
	DBStd	10.38	5.98	5.76	6.07	4.36	4.22	3.59	3.08	3.07	4.09	4.54	4.53	4.80		
	HDD10.0	2317	521	473	381	182	41	1	0	0	4	82	230	403		
	HDD18.3	4489	779	707	637	426	239	88	24	34	116	302	477	662		
	CDD10.0	1009	0	0	2	6	63	182	292	266	154	41	3	0		
	CDD18.3	139	0	0	0	0	3	19	57	41	16	2	0	0		
	CDH23.3	1233	0	0	4	5	66	188	518	327	112	13	0	0		
	CDH26.7	268	0	0	1	1	15	43	124	63	19	1	0	0		
(14) Wind	WSAvg	3.1	3.3	3.6	3.3	3.6	2.9	2.6	2.5	2.6	2.6	3.2	3.4	3.5		
(15) Precipitation	PrecAvg	745	42	32	44	61	68	66	77	81	93	69	63	51		
	PrecMax	1028	99	104	133	155	185	190	195	192	239	153	136	144		
	PrecMin	339	7	5	2	7	7	10	7	4	7	7	11	7		
	PrecStd	139	20	20	30	33	34	36	43	42	52	36	31	31		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	8.5	17.0	22.5	28.9	30.5	32.2	31.1	29.5	25.3	19.1	11.6		
		MCWB	5.2	6.0	11.6	13.5	18.3	21.1	22.3	22.0	21.5	17.5	13.0	10.4		
	2%	DB	4.4	4.1	11.8	17.2	25.0	27.7	29.7	28.2	26.4	22.0	13.9	6.4		
		MCWB	2.5	2.3	8.2	10.5	16.8	20.0	21.1	20.8	19.3	17.1	10.2	4.4		
	5%	DB	2.6	2.3	8.0	13.7	22.0	25.6	27.8	26.8	24.1	18.2	11.3	4.7		
		MCWB	1.4	0.9	4.7	8.0	14.9	18.7	20.3	19.8	18.7	14.9	8.6	3.4		
	10%	DB	1.1	0.9	5.9	11.2	18.9	23.0	26.3	25.1	22.5	16.0	9.0	3.0		
		MCWB	0.4	-0.3	3.0	6.6	13.6	17.2	19.6	18.9	17.8	12.9	7.0	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.5	12.9	14.9	20.3	22.8	24.0	23.4	22.5	20.0	13.7	10.2		
		MCDB	6.9	8.6	16.0	20.9	25.5	27.7	29.3	28.3	26.9	22.7	17.4	11.4		
	2%	WB	2.9	2.7	8.4	11.2	18.4	21.1	22.6	22.2	21.0	17.5	10.9	5.2		
		MCDB	4.0	4.0	11.2	15.8	23.1	26.1	27.3	26.5	24.4	20.1	13.0	6.0		
	5%	WB	1.5	1.0	5.2	8.9	16.2	19.8	21.6	21.2	19.9	15.3	8.9	3.6		
		MCDB	2.4	2.0	8.0	13.0	21.0	24.2	26.0	24.8	23.4	18.3	11.0	4.5		
	10%	WB	0.3	-0.1	3.2	7.1	14.2	18.5	20.6	20.3	18.7	13.2	7.3	2.1		
		MCDB	1.1	0.7	5.2	10.4	18.2	22.2	24.8	23.6	21.7	15.8	8.7	3.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.4	9.5	9.8	12.1	11.8	11.6	11.0	10.9	8.6	6.8	5.7		
		MCDBR	7.0	8.2	12.6	15.0	17.0	14.4	13.4	12.7	12.8	12.7	11.0	7.4		
	5% WB	MCWBR	6.0	6.9	8.9	9.2	9.4	7.7	6.7	6.5	7.0	8.2	7.9	6.5		
		MCWBR	6.7	8.0	11.9	14.0	14.6	12.7	11.6	10.6	11.2	11.5	10.1	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.293	0.317	0.358	0.376	0.395	0.392	0.384	0.369	0.335	0.303	0.280		
		taud	2.298	2.337	2.395	2.338	2.364	2.343	2.371	2.389	2.421	2.498	2.519	2.446		
	Ebn at Noon		793	888	915	900	892	873	871	864	848	833	805	796		
		Edh at Noon	79	92	101	118	119	122	117	111	99	79	65	62		
(44) All-Sky Solar Radiation	RadAvg		1.14	2.07	3.50	4.61	5.65	6.13	6.17	5.23	3.98	2.31	1.22	0.84		
	RadStd		0.13	0.21	0.36	0.48	0.48	0.44	0.24	0.33	0.32	0.27	0.17	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page